DEEPAN R L

Software Developer

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PROFESSIONAL SUMMARY

Python Developer with 3 years of experience specializing in computer vision, image processing, and industrial automation systems. Proven expertise in developing vision inspection solutions using OpenCV, Django, and Flask. Hands-on experience building 10 + industrial inspection applications, real-time detection systems, and safety monitoring.

TECHNICAL SKILLS

Programming Languages: Python
Frameworks & Libraries: Django, Flask, OpenCV, NumPy, Matplotlib
Build & Version Control : Git , GitHub, Pyinstaller, Inno Setup
Computer Vision: Image Processing, Object Detection, OCR, Crack Detection,Depth Measurement
Communication & Networking: TCP/IP Communication, Socket Programming ,RTSP Video Streaming
Camera & Hardware: SICK Visionary-S, SICK Ruler, Pico Cam , Sensing Cam
Development Tools: Vs code
Domain Knowledge: Industrial Automation, Vision Inspection, Safety Monitoring, Tire Inspection

PROFESSIONAL EXPERIENCE

Python Developer Aug 2023 – Present

ERL Sensors and AutomationChennai, India

- Developed 8+ computer vision applications for industrial inspection including circle center detection, bolt detection, crack detection, OCR-based part identification, person detection, tire inspection, image-to-graph visualization, bulge detection, and multi-camera detection systems
- **Bolt Detection System:** Implemented angle measurement for bolt detection using SICK Ruler, achieving accurate orientation analysis for quality control applications
- **Crack Detection:** Created crack detection system using Pico Cam with optimized image processing algorithms for surface defect identification in manufacturing environments
- **OCR Part Identification:** Developed OCR-based part identification system enabling automated recognition and classification of components in production lines
- **Safety Monitoring:** Built person detection system for real-time safety monitoring in industrial environments, reducing safety incidents through automated alerts
- **Tire Inspection:** Engineered comprehensive tire inspection systems with multi-parameter defect detection and automated reporting
- **Human Detection:** Developed a real-time human detection system using Raspberry Pi integrated with Hailo AI accelerator for efficient edge-based safety monitoring.
- **3D Palletization Application:** Built a 3D palletization solution using depth data and computer vision to optimize object placement, stacking, and space utilization in industrial environments.
- **RTSP Streaming Application:** Implemented real-time video streaming using RTSP, FFmpeg, and MediaMTX. Integrated camera feeds with OpenCV for live video processing and network-based monitoring.
- **Glass Damage Detection:** Developed a 3D glass damage detection system for **First Solar** using a **SICK Ruler 3D camera** for real-time defect inspection and quality analysis.

AI & Machine Learning Projects

- Custom YOLO Object Detection: Trained and deployed custom YOLO models for industrial object detection and defect identification. Performed dataset collection, annotation, model training, validation, and deployment.
- **Human Detection System:** Implemented real-time human detection using Raspberry Pi and Hailo AI accelerator for edge-based safety applications.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering	2019 – 2022
Tagore Institute of Engineering and Technology	CGPA: 7.99/10 (79.9%)

PORTFOLIO & LINKS

- Portfolio: deepanportfolio.onrender.com/
- GitHub : github.com/gididdeepan?tab=repositories